Slot engine: how the new system fits together

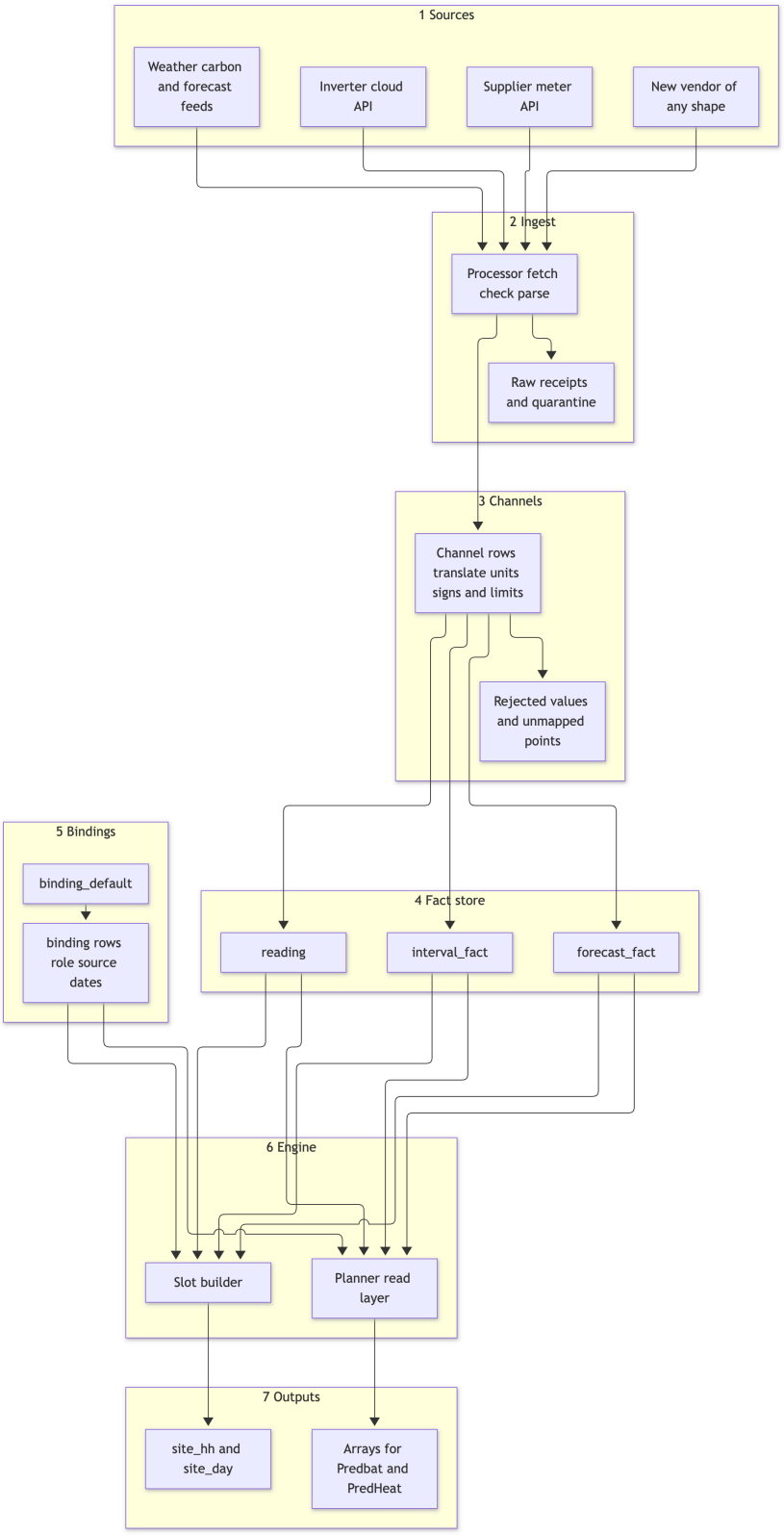
This page draws the provider-agnostic slot system as it is built today, in Mermaid. Each diagram marks what exists (solid, green) and what is planned (dashed, grey). The reasoning behind the design is in `docs/slot-engine-solution.md`; the root causes it answers are in `docs/resilient-slot-engine.md`.

The idea in one line. The slot builder never asks for a vendor. It asks for a *role* ("the inverter's 5-minute samples"), and a dated *binding* says which source fills that role for this home. A vendor's own units and signs are handled by *channel* rows when its data is written, so a new provider is rows, not engine code.

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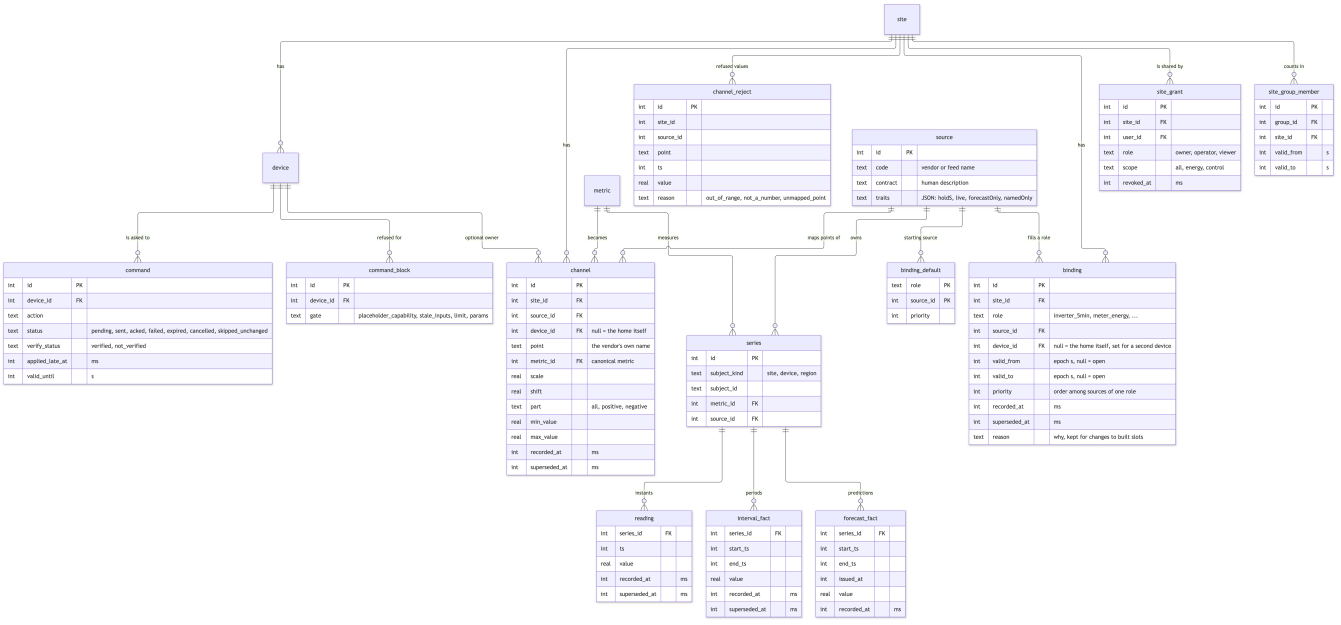
1. Overview



How to read it: data flows top to bottom. Vendors differ only in step 2. Steps 3 and 5 are rows in tables, not code, so adding, switching or removing a provider changes rows only. Both engine parts in step 6 read the same bindings, so the slot builder and the planner agree on which source a slot used (within 0.1 percent in the tests).

2. Data model

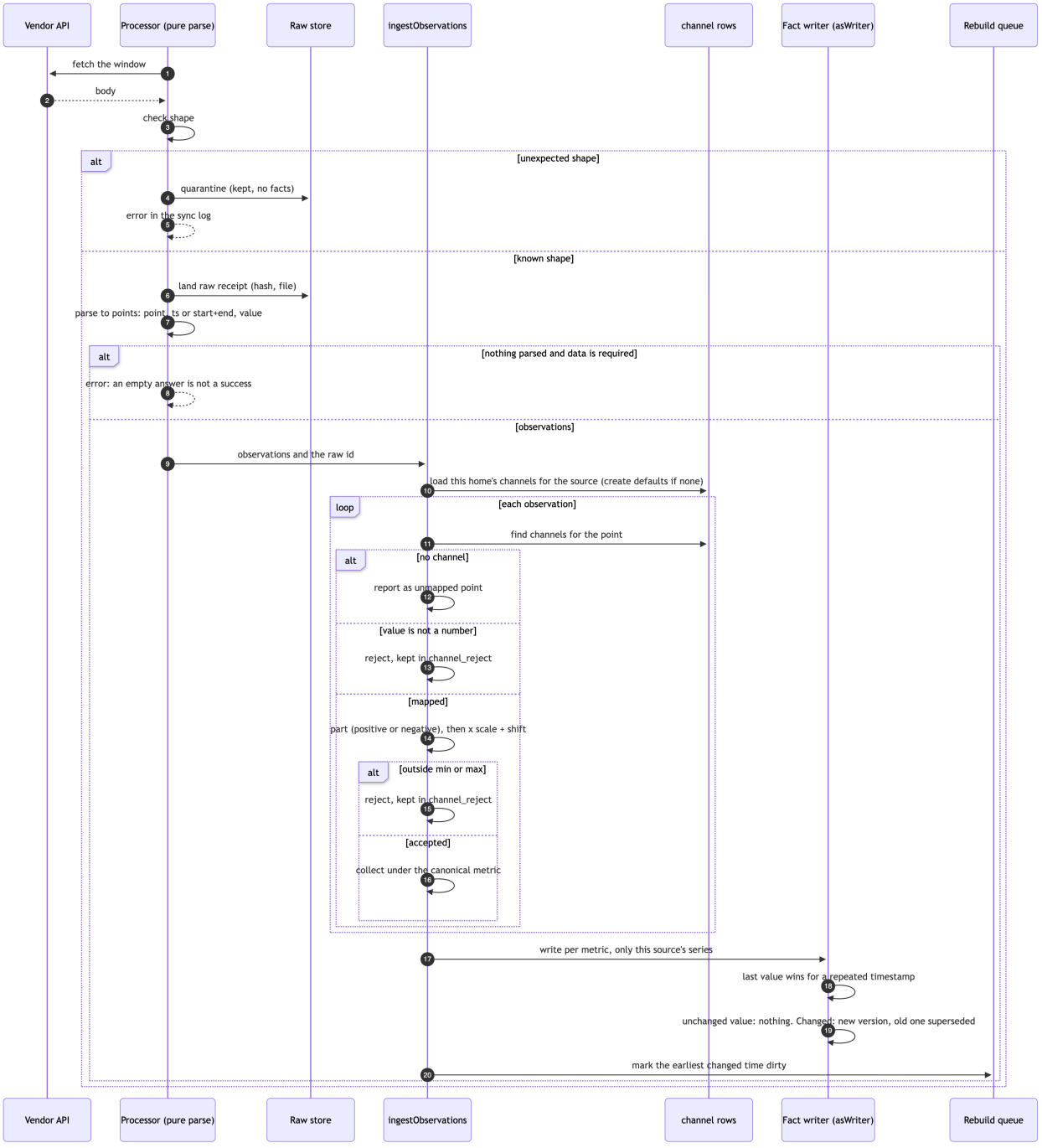
Only the tables this design adds or relies on, with the control and access tables in brief (their full definitions are in `docs/control-readback.md` and `docs/groups-and-access.md`). `source.traits` is a new column; `channel`, `channel_reject`, `binding` and `binding_default` are new tables (deltas 061 to 066).



3. Writing data in

Every vendor takes the same path. The processor knows the vendor's shape; the channels know the vendor's units and signs; the fact writer knows nothing about vendors and refuses writes outside the declared source.

Observations come in six kinds: an instant, a period, a forecast issue, a **counter** (a cumulative total such as a plug's energy meter, turned into energy per period by difference, with resets, wraps and backwards steps handled), an **event** (a 0 or 1 period), and the control kinds **state** and **ack**. The kinds are described in `docs/counters-events-roles.md` and `docs/control-readback.md`.



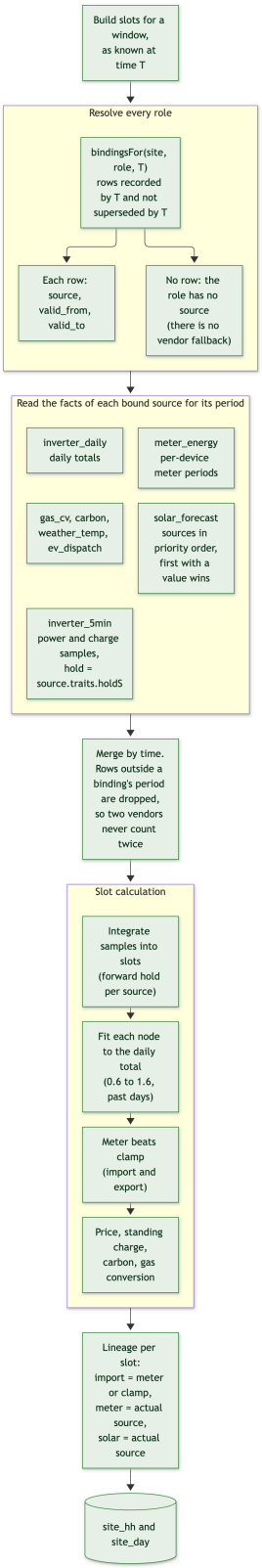
What the channel row does, by example (the same inverter reported two ways):

Vendor point	Vendor value	Channel (metric, part, scale)	Stored fact
`Grid` (signed W, + = import)	+500	`grid_import_w`, positive, 1	500
`Grid`	-300	`grid_import_w`, positive, 1	0
`Grid`	-300	`grid_export_w`, negative, 1	300
`grid_kw` (kW, a second vendor)	-0.3	`grid_export_w`, negative, 1000	300

Vendor point	Vendor value	Channel (metric, part, scale)	Stored fact
`pv_kw`	500000	`solar_w`, scale 1000, max 60000	rejected, kept in `channel_reject`
`firmware_temp`	41	none	reported as unmapped, not stored

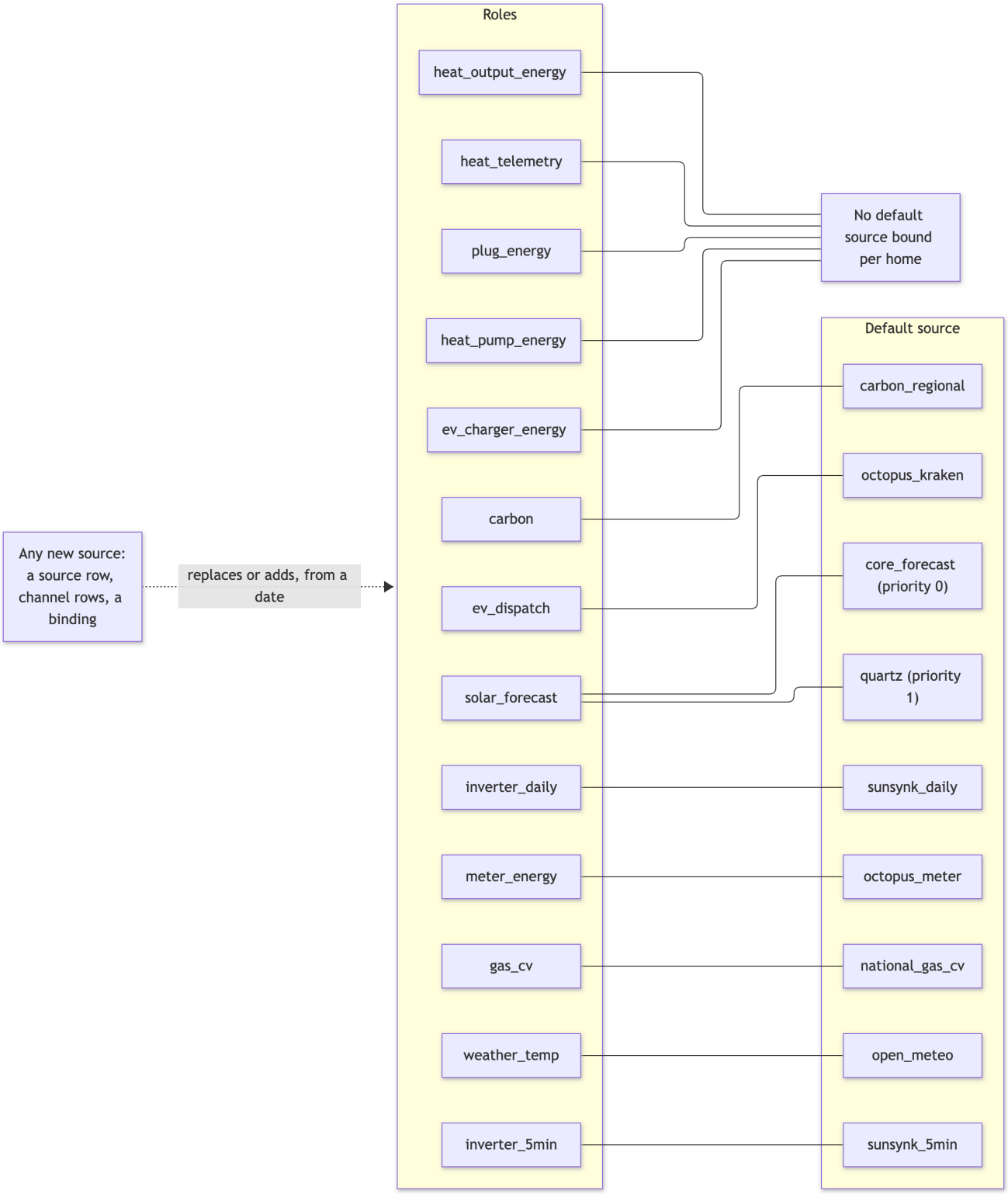
4. Reading data out

The builder and the planner read layer take their inputs from the same bindings, so they cannot disagree about which source a slot used.



5. Roles

A role is what the engine asks for. A source fills it. The defaults below are what a new home starts with (`binding_default`); a home can bind anything else.



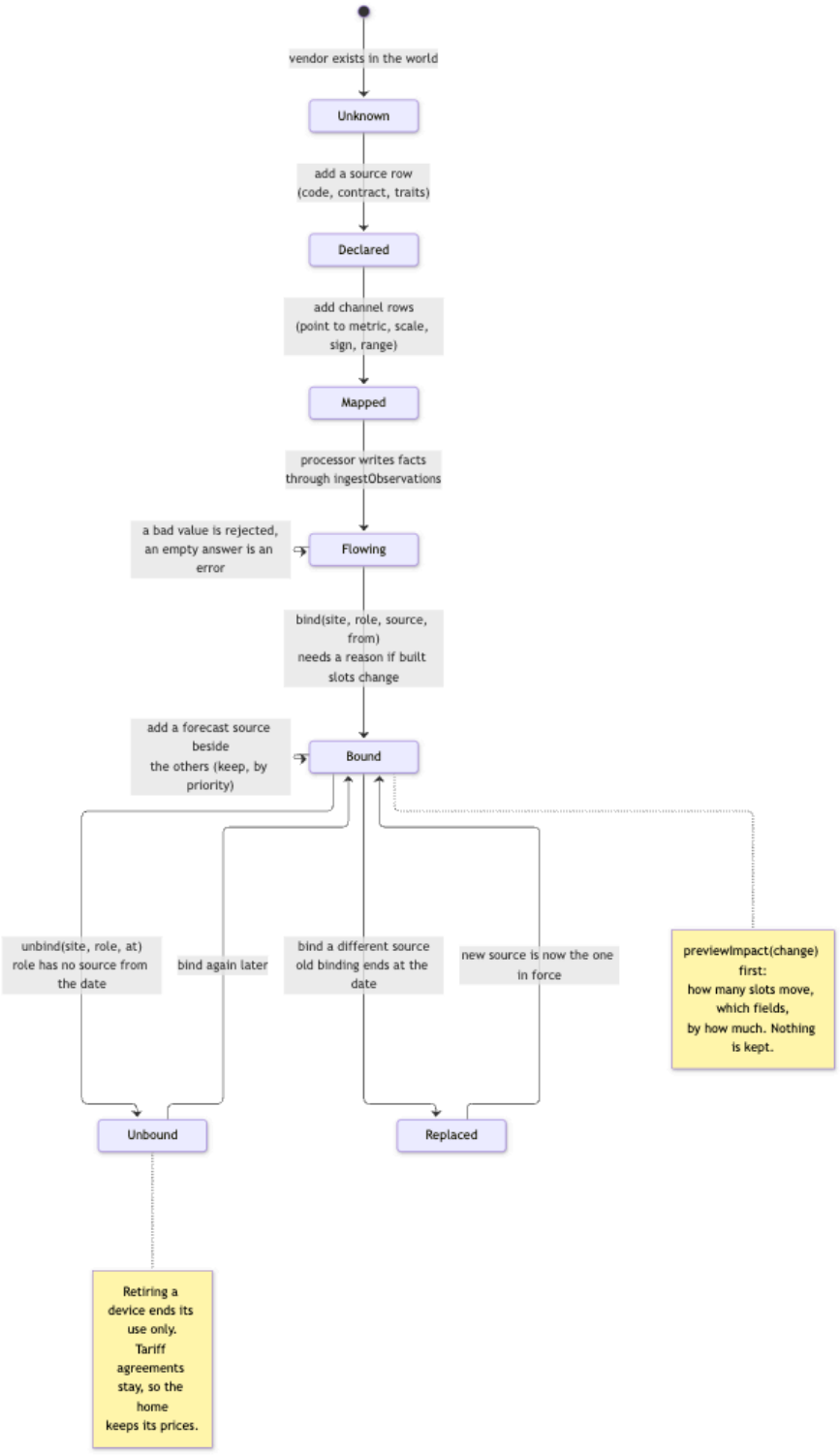
Role	What the builder uses it for	Stored as
`inverter_5min`	Solar, load, grid, battery power and charge level, integrated into slots	`reading`, site subject
`inverter_daily`	The inverter's daily totals, used to fit the integrated slots	`interval_fact`, site subject
`meter_energy`	The supplier meter's import, export and gas per period (official figure)	`interval_fact`, device subject

Role	What the builder uses it for	Stored as
`gas_cv`	Gas calorific value, to convert m3 to kWh	`interval_fact`, region subject
`carbon`	Grid carbon intensity per period	`interval_fact`, region subject
`weather_temp`	Outdoor temperature	`reading`, site subject
`ev_dispatch`	Smart-charge dispatch periods (the daily allowance)	`interval_fact`, site subject
`solar_forecast`	The forecast of record per slot, our own first, then Quartz	`forecast_fact`, site subject
`heat_pump_energy`	Electricity into a heat pump (read by the heating build)	`reading` or counter, device subject
`heat_output_energy`	Heat delivered by a heat pump or boiler (read by the heating build)	counter or `interval_fact`, device subject
`heat_telemetry`	Heating temperatures and on or off states (read by the heating build)	`reading`, site or device subject
`ev_charger_energy`	Energy delivered to an EV by a charger (defined, no consumer yet)	counter or `interval_fact`, device subject
`plug_energy`	A smart plug's energy, per device (defined, no consumer yet)	counter, device subject

The five heating and IoT roles have no default source: a home binds one when it has the device. Roles are read from one list (`ROLES` in `src/core/inputs.js`), so the Providers screen shows new ones automatically.

6. Provider lifecycle

Everything below is a row change, done while the system runs. Slots before a change never move.



Each inverter binding is a *unit*: a source, optionally a device, for a period. Units that follow one another take turns (a vendor switch). Units that overlap in time are two devices of one kind (two inverters, two batteries) and their figures are **added**: each is integrated on its own with its own hold, the slot energies are summed, a slot is covered only as far as its least covered unit, and a day's counter total is the sum of the units' totals (unknown if one is missing). Charge level is the average of the batteries weighted by size (the device's `battery_kwh`). `solar_forecast` is the one role that has several sources at once by priority instead of adding them.

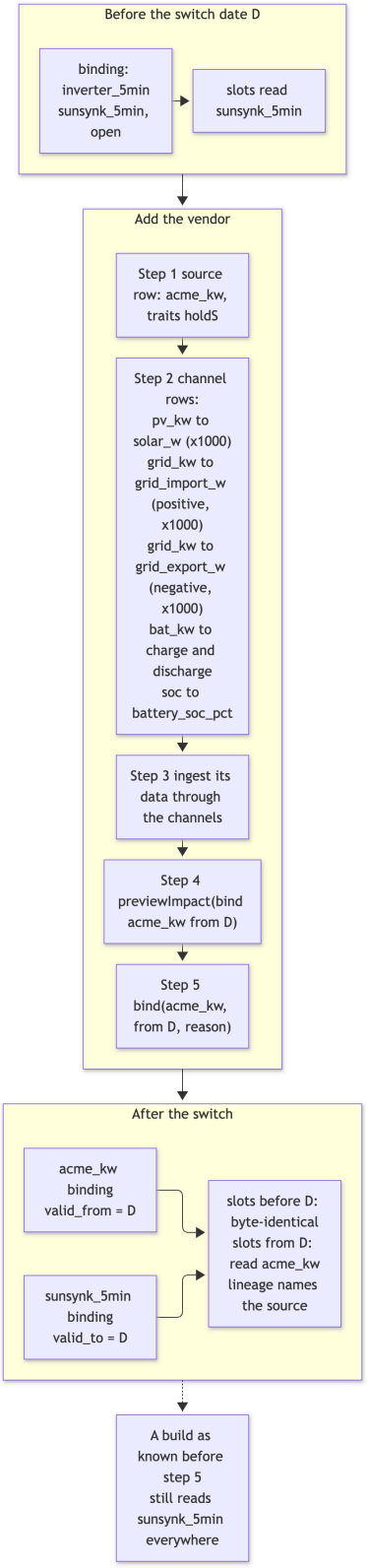
Safety rules that go with it

Rule	Where it is enforced
A change that alters built slots needs a written reason, kept on the binding row	`inputs.js` (`bind`, `unbind`)
A processor can only write series of its own source	`facts.js` (`asWriter`)
A new source or metric row is found without a restart	`db.js` (lookup on a cache miss)
A "preview" runs the change in a savepoint and rolls it back	`impact.js` (`previewImpact`)
A build "as known before" a change still sees the old binding	binding `recorded_at` and `superseded_at`, `bindingsFor(..., asOf)`

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7. A vendor switch, worked through

Tested on the real home's data (`test/runtime-provider.test.js`, `test/channels.test.js`): a second inverter vendor reports in kW with one signed grid figure, is wired in as rows, and is bound from a date.

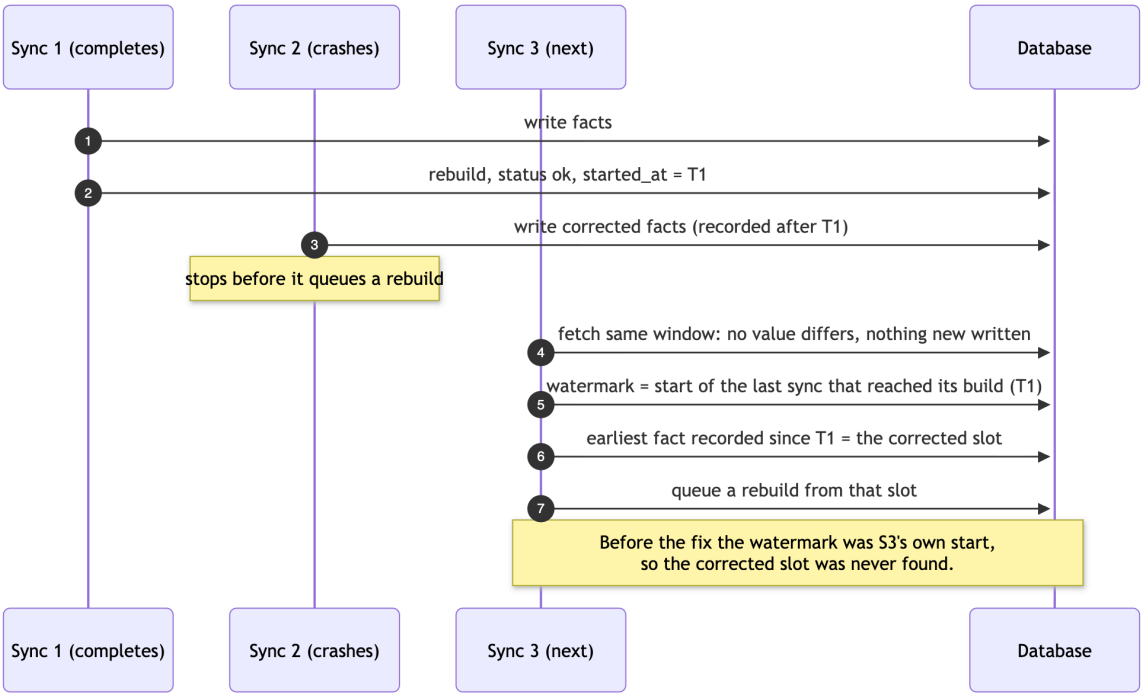


Result of that test: slots from D equal the real inverter's within 1e-6 on solar, home use, import, export, battery charge and discharge, and total cost. The planner read agrees with the builder within 0.1% across the switch.

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8. Crash safety

The old failure: a sync wrote corrected facts, stopped before its rebuild, and the next sync fetched the same window, found nothing changed, and never rebuilt the day.

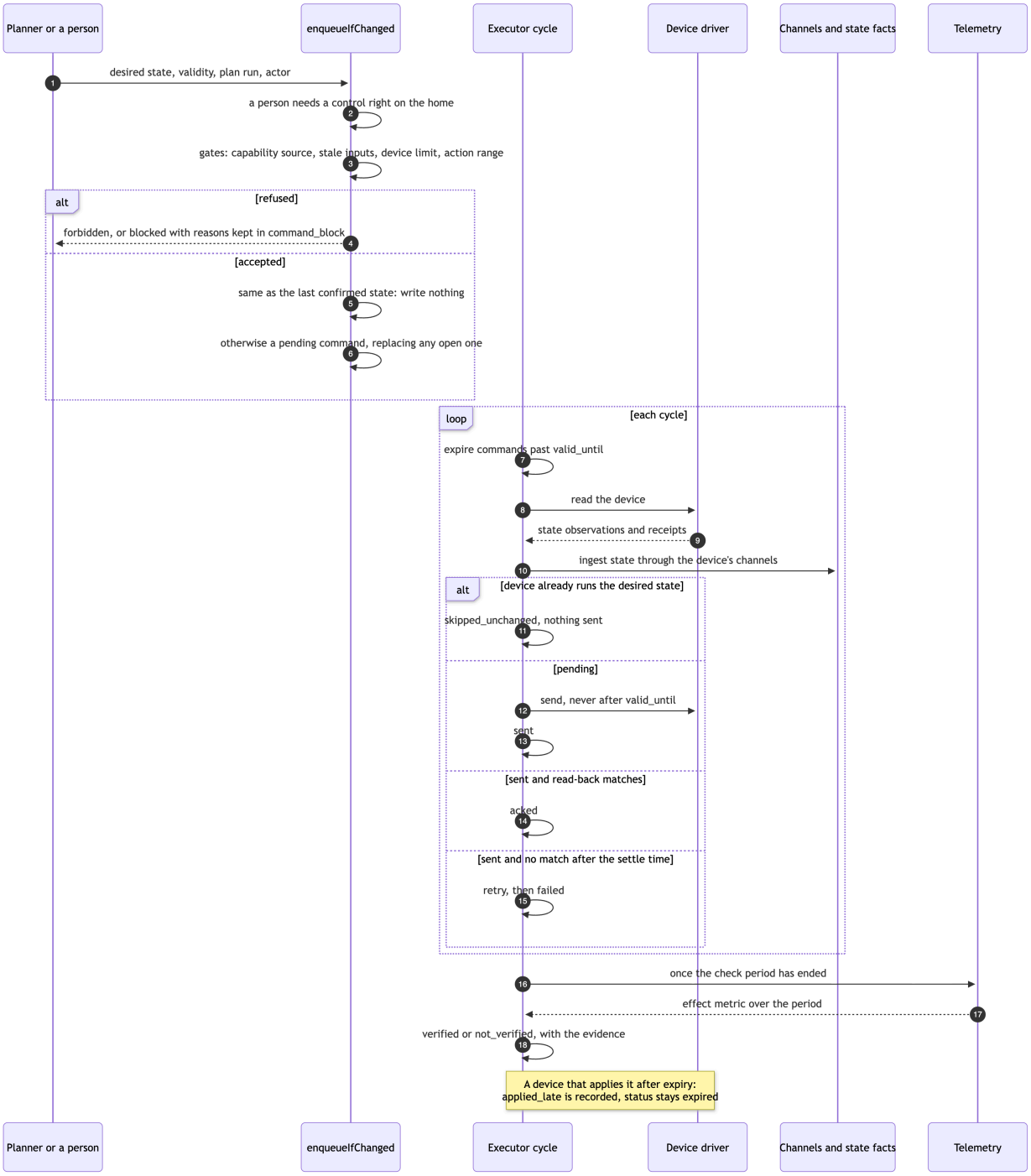


Also in this stage: a batch with the same timestamp twice keeps the last value (it used to break the unique key and abort the whole batch), and `requireData` turns an empty answer into an error the sync log shows.

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9. Control: a command with read-back

Writing to a device uses the same channels as reading from it. A device's settings come back as `state` observations through the same ingest path, a receipt is an `ack` that is never stored as a fact, and a command counts as done only when the **device** reports the desired state. Full detail: `docs/control-readback.md`.

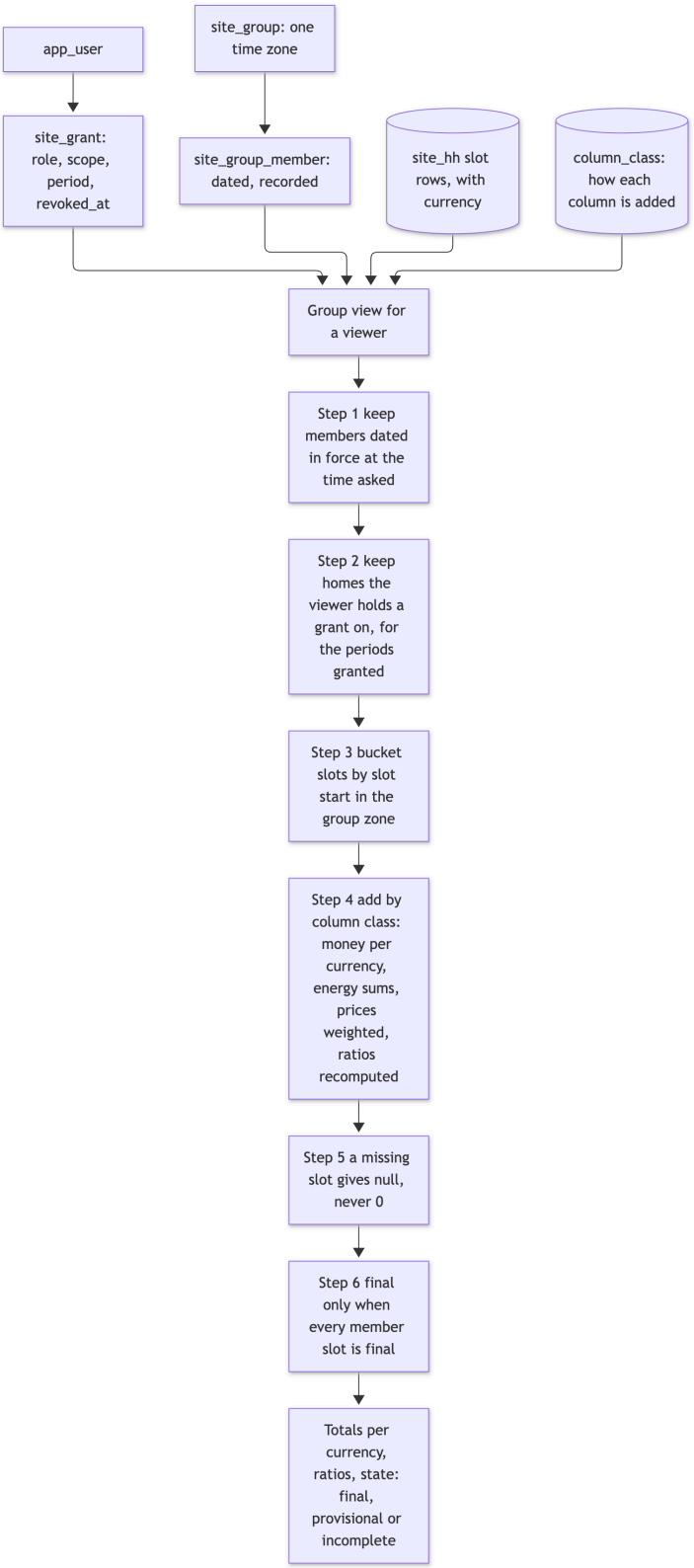


Gate or rule	What it stops	Kept where
Control right	A viewer, a stranger or a revoked user commanding a home	error `forbidden`, nothing written
Placeholder capability	A command for an action whose support was only stated, not confirmed	`command_block`
Stale inputs	A non-demand action planned on stale data	`command_block`
Device limit	A value outside the installer's limit for that device	`command_block`
Action range	A value outside the action's own parameter range	`command_block`

Gate or rule	What it stops	Kept where
Read-back decides	Marking a command done on the database's belief	`acked` only when the device reports it

10. Groups and access

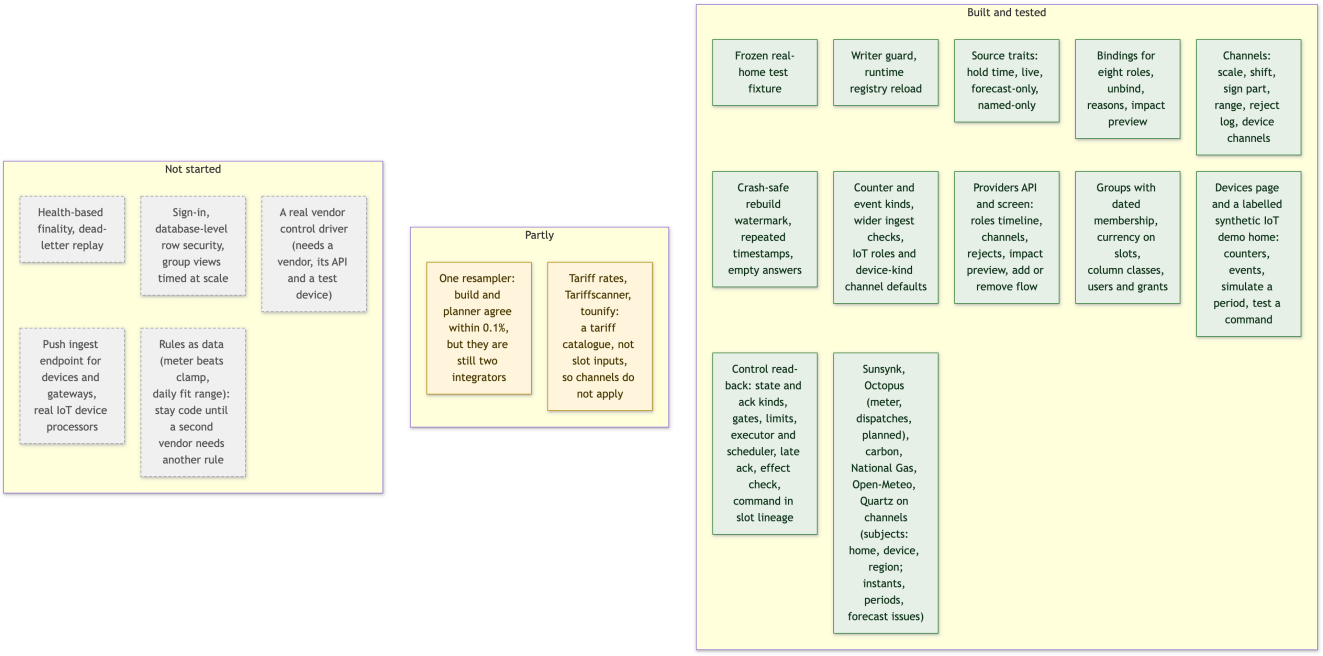
A group is a dated set of homes. Its views are built from the homes' slot rows, never from daily totals, and who may see or command a home is a grant, not a tenant. Full detail: `docs/groups-and-access.md`.



Question	Answer in this design
A group has GBP and EUR homes	Two money totals; no exchange rate exists
A home leaves on day X	Counted before X and not after; an earlier report re-runs identically
A member the viewer has no grant on	Left out completely: no count, no gap marker
One member slot is missing	The column is null with a `partial` figure; never 0

Question	Answer in this design
Homes with different slot lengths	Default bucket is the longest slot; a shorter bucket is refused, not split
A viewer with scope energy	Reads energy and cost; cannot command

11. What is built and what is not



Where each piece lives

Piece	File
Roles, bindings, `bind`, `unbind`, hold lookup	`src/core/inputs.js`
Impact preview	`src/core/impact.js`
Channels, `ingestObservations`, `setDeviceChannel`	`src/core/channels.js`
Writer guard, fact writers, rebuild queue	`src/core/facts.js`
Slot builder	`src/core/build.js`
Planner read layer	`src/core/planner-read.js`
Sync watermark	`src/jobs/sync.js` (`sinceWatermark`)

Piece	File
Source definitions, default channels, traits	`src/db/registry.js`
Default bindings seed	`src/db/seed/bindings.js`
Schema	`src/db/deltas/061` to `066`
Processor template	`src/adapters/_template.js`
Control (stage 4)	`src/core/executor.js`, `src/core/control.js`, `src/core/commands.js`, `src/db/deltas/067-control.sql`; see `docs/control-readback.md`
Counters, events and IoT roles	`src/core/counters.js`, `src/core/channels.js`, `src/core/heat.js`, `src/db/deltas/072-channel-kinds.sql`; see `docs/counters-events-roles.md`
Providers API and screen	`src/core/onboarding.js`, routes in `src/server.js`, `public/providers.html`, `public/js/pages/providers.js`; see `docs/onboarding-api.md`
Devices page and synthetic IoT demo	`src/core/devices-view.js`, `src/core/synthetic-iot.js`, `public/devices.html`, `public/js/pages/devices.js`, `npm run demo:iot`; see `docs/devices-page.md`
Groups and access (stage 5)	`src/core/groups.js`, `src/core/access.js`, `src/db/deltas/070-groups.sql`; see `docs/groups-and-access.md`
Tests	`test/runtime-provider.test.js`, `test/channels.test.js`, `test/ingest-durability.test.js`, `test/sunsynk-channels.test.js`, `test/octopus-channels.test.js`, `test/channels-subjects.test.js`, `test/multi-device.test.js`

Reading these diagrams

Mermaid renders in GitHub, GitLab, VS Code (Markdown preview with the Mermaid extension) and most documentation tools. If a diagram shows as plain text, open the file in one of those.